RESEARCH STATEMENT

The category $\mathcal{M}\mathrm{fld}$ of smooth finite-dimensional manifolds is the basic setting for differential geometry. It is notoriously rigid, with nice objects but poor properties: subsets and quotients of, and mapping spaces between, manifolds are rarely themselves manifolds. These deficiencies obstruct several important results, such as

- Symplectic reduction: given a Hamiltonian action of a Lie group G on a symplectic manifold (M, ω), with a momentum map $\mu\colon M \to \mathfrak{g}^*$, the reduced space $M_{\mathrm{red}} := \mu^{-1}(0)/G$ is a symplectic manifold if and only if the G-action is free and proper.
- Infinite-dimensional Lie theory: if M is a non-compact manifold, there is no (infinite-dimensional) manifold structure for which the Lie algebra of $\mathrm{Diff}(M)$ is isomorphic to the Lie algebra of vector fields $\mathfrak{X}(M)$.

Diffeological spaces, introduced by Souriau [Sou80], generalize manifolds. In sophisticated terms, a diffeological space is a concrete sheaf on the site $\mathcal{O}\mathrm{pen}$ of open subsets of Cartesian spaces. In grounded terms, a diffeology on a set X is a collection of maps $\mathscr{D} = \{U \xrightarrow{p} X \mid U \in \mathcal{O}\mathrm{pen}\}$ which contains all constant maps and satisfies the sheaf axioms. The category $\mathcal{D}\mathrm{flg}$ of diffeological spaces is geometric, and is closed under taking subsets, quotients, and mapping spaces.[(1)] We could now ask, for instance, "is M_{red} always a symplectic diffeological space, with symplectic form induced by ω?" or "is $\mathrm{Lie}(\mathrm{Diff}(M)) \cong \mathfrak{X}(M)$, and what is Lie?"

Of course, there are other generalizations of smooth manifolds that can address such questions. Taking a cue from algebraic geometry and moving to higher categorical settings, one can use orbifolds, and more generally differentiable stacks, to model quotients of manifolds. Taking a cue from functional analysis, one may consider infinite-dimensional Banach, Fréchet, or convenient[(2)] manifolds to handle mapping spaces.

My research combines the infinite-dimensional, higher geometric, and diffeological approaches to generalize those qualified theorems in $\mathcal{M}\mathrm{fld}$, or its infinite-dimensional incarnations, beyond their obstructions. The results and forecasts of these efforts are captured by the specific examples in the sequel.

1. Infinite-dimensional Lie theory

Let us begin with the well-known functor

$$\mathrm{Lie}\colon \{\text{Lie groups}\} \to \{\text{Lie algebras}\}, \quad G \mapsto T_eG.$$

Defining Lie traditionally requires the tangent functor $\hat{T}\colon \mathcal{M}\mathrm{fld} \to \mathcal{M}\mathrm{fld}$, and a Lie algebra structure on $\mathfrak{X}(M)$ that is typically given by identifying it with the algebra of derivations on $C^\infty(M, \mathbb{R})$. But this identification does not generalize to diffeological spaces; many spaces that we require to have non-trivial Lie algebras, such as irrational tori $\mathbb{R}/(\mathbb{Z} + \alpha\mathbb{Z})$ for irrational α, carry only trivial derivations. Thus alternatively, we realize that we can recover the Lie bracket on $\mathfrak{X}(M)$ using the entire *tangent structure* on M, in the sense of Rosický [Ros84]: this is the data of T, together with

(1) It is a quasitopos.

(2) In the sense of Kriegl and Michor [KM97].

the natural transformations

the projection	$\pi\colon T \to 1$
the zero-section	$0\colon 1 \to T$
the addition	$+\colon T_2 \to T$, where $T_2 M \coloneqq TM \times_\pi TM$
the vertical lift	$\lambda\colon T \to T^2$
the canonical flip	$\tau\colon T^2 \to T^2$,

satisfying a list of axioms. Cockett and Cruttwell [CC15] have spearheaded modern research into tangent categories. For $\mathcal{D}\mathrm{flg}$, we begin by choosing the tangent functor

$$T \coloneqq \mathrm{Lan}_y(y \circ \hat{T}),$$

the left Kan extension of $\hat{T}\colon \mathcal{M}\mathrm{fld} \to \mathcal{M}\mathrm{fld}$ along the embedding $y\colon \mathcal{M}\mathrm{fld} \to \mathcal{D}\mathrm{flg}$. In pioneering work towards diffeological Lie theory, Blohmann isolated a full subcategory $\mathcal{E}\mathrm{last}$ of $\mathcal{D}\mathrm{flg}$, whose objects are called *elastic* spaces, for which:

Theorem (Blohmann [Blo24])**.** *The category $\mathcal{E}last$, equipped with T and (more-or-less)*[(3)] *the left Kan extensions of all the tangent structure data on $\mathcal{M}fld$, is a tangent category.*

As a consequence, elastic diffeological groups G have elastic Lie algebras $\mathfrak{g} = T_e G$, with Lie algebra structure induced by the Lie bracket on left-invariant vector fields on G. The article [Blo24] announces several results; with Christian Blohmann, we are preparing the longer manuscript containing all the proofs.

Question 1. For the functor

$$\mathrm{Lie}\colon \{\text{elastic groups}\} \to \{\text{elastic Lie algebras}\},$$

- Does every elastic Lie algebra integrate to an elastic group?
- Are "simply-connected" integrations universal?
- How do Lie algebra morphisms lift to elastic group morphisms?

This instantiates a program. In joint work in-progress with Christian Blohmann, we are pursuing the first point. We have proved:

Theorem 2. *If M is compact, the group $\mathrm{Diff}(M)$ is elastic, with Lie algebra isomorphic to $\mathfrak{X}(M)$.*

Our goal is to show the same is true for M non-compact. This lies outside the realm of infinite-dimensional manifolds, where $\mathrm{Diff}(M)$ is modelled on the space of vector fields of compact support. We have thus far showed that $\mathfrak{X}(\mathbb{R}^n)$ is a quotient of $T_{\mathrm{id}}\,\mathrm{Diff}(\mathbb{R}^n)$.

Even within the realm of infinite-dimensional geometry, elastic groups clarify and extend Lie theory. In finite-dimensions, we have Lie's celebrated third theorem.

Theorem (Lie)**.** *For every finite-dimensional Lie algebra $\mathfrak{g}$, there is a Lie group G with $\mathrm{Lie}(G) = \mathfrak{g}$.*

Cartan constructed G by integrating central extensions of Lie algebras. Let $\mathfrak{z}$ denote the center of $\mathfrak{g}$, and let $\mathfrak{g}_{\mathrm{ad}}$ denote $\mathfrak{g}/\mathfrak{z}$. The latter integrates to a simply-connected Lie group G_{ad}, since it acts faithfully on $\mathfrak{g}$. The former integrates to $(\mathfrak{z}, +)$, or its quotient by any topologically discrete subgroup Π. We seek G fitting into the diagram:

$$\begin{array}{ccccccccc} \{e\} & \longrightarrow & \mathfrak{z}/\Pi & \longrightarrow & G & \longrightarrow & G_{\mathrm{ad}} & \longrightarrow & \{e\} \\ \downarrow{\scriptstyle \mathrm{Lie}} & & \downarrow{\scriptstyle \mathrm{Lie}} & & \downarrow{\scriptstyle \mathrm{Lie}} & & \downarrow{\scriptstyle \mathrm{Lie}} & & \downarrow{\scriptstyle \mathrm{Lie}} \\ 0 & \longrightarrow & \mathfrak{z} & \longrightarrow & \mathfrak{g} & \longrightarrow & \mathfrak{g}_{\mathrm{ad}} & \longrightarrow & 0. \end{array} \tag{1}$$

(3) The actual tangent structure data are not all Kan extensions.

The extension of Lie algebras is completely determined by the Lie algebra 2-cocycle $\omega\colon \mathfrak{g}_{\mathrm{ad}}\times\mathfrak{g}_{\mathrm{ad}} \to \mathfrak{z}$, which extracts the $\mathfrak{z}$-component of the Lie bracket. This, in turn, determines the *group of periods* $\Pi_\omega \leq \mathfrak{z}$, which arises as the image of $\pi_2(G_{\mathrm{ad}})$ under a group homomorphism. If Π_ω is topologically discrete, then we may construct a G that completes the diagram (1), with $\Pi = \Pi_\omega$. When $\mathfrak{g}$ is finite-dimensional, the group G_{ad} is a Lie group, and so $\pi_2(G_{\mathrm{ad}})$, hence also Π_ω, is trivial. Thus we have integrated $\mathfrak{g}$ without leaving $\mathcal{M}$fld.

This fails in infinite dimensions: there are examples Banach Lie groups for which Π_ω is not topologically discrete (the first was given in [EK64]). However, this is not an obstruction for diffeology. I have proved[(4)]

Theorem 3 ([Miy25])**.** *Let $\mathfrak{g}$ be a Banach Lie algebra, whose center $\mathfrak{z}$ is complemented. The integration of the central extension $\mathfrak{z} \to \mathfrak{g} \to \mathfrak{g}_{ad}$ as indicated by (1) exists in $\mathcal{E}$lst if $\pi_2(G_{ad})$ is countable.* A fortiori, *$\mathfrak{g}$ integrates to a diffeological group.*

Remark 4. Implicit in the statement of this theorem is the fact, also proved in [Miy25], that the tangent category of Banach (more generally Fréchet or convenient) manifolds embeds into $\mathcal{E}$lst.

The countability of $\pi_2(G_{\mathrm{ad}})$ ensures that Π_ω is diffeologically discrete; the quotients $\mathfrak{z}/\Pi_\omega$ are infinite-dimensional irrational tori. Integration to stacks, without an assumption on $\pi_2(G_{\mathrm{ad}})$, was carried out in [Woc11; WZ16]. Formalizing the connection between the stacky and diffeological integrations of $\mathfrak{g}$, and resolving the diffeological integration of $\mathfrak{g}$ without assumption on $\pi_2(G_{\mathrm{ad}})$, constitute two avenues for future research. More concretely, the question of which Banach Lie algebras have diffeologically discrete $\pi_2(G_{\mathrm{ad}})$ is also open; Banach Lie algebras with non-discrete $\pi_2(G_{\mathrm{ad}})$ would be excellent test-cases for the aforementioned comparison.

2. Integrating Lie algebroids and singular foliations

Lie groupoids and Lie algebroids are "horizontal" generalizations of Lie groups and Lie algebras. Examples include Lie group and Lie algebra bundles. We again have a Lie functor:

$$\mathrm{Lie} := \{\text{Lie groupoids}\} \to \{\text{Lie algebroids}\}, \quad \mathcal{G} \mapsto A.$$

Non-integrable Lie algebroids were found by [AM85]. Crainic and Fernandes [CF03] gave necessary and sufficient conditions for integrability. When A is transitive over a simply-connected base, there is an associated "period group" $\Pi(A)$ identifying with a proper subgroup of $(\mathbb{R}, +)$, and

Theorem (Crainic-Fernandes)**.** *A is integrable if and only if $\mathbb{R}/\Pi(A)$ is a manifold.*

Once again, $\mathbb{R}/\Pi(A)$ is an elastic diffeological space, so we pose the question:

Question 5. Does every transitive Lie algebroid integrate to an elastic diffeological groupoid?

The answer is likely to be "yes," because the troublesome quotients appearing in the attempted integration of a transitive Lie algebroid are those of the form $\mathbb{R}/\Pi(A)$, which are all elastic. Therefore the question may be colourably called a conjecture. But we have begged the genuine question:

Question 6. Is there a Lie functor

$$\mathrm{Lie} : \{\text{elastic Lie groupoids}\} \to \{\text{elastic Lie algebroids}\}?$$

What constitutes the right-hand side?

This instantiates another program. Blohmann and Aintablian's work [AB25] proposes a Lie functor. With other tools, Villatoro [Vil26] introduced a Lie functor for yet another class of diffeological groupoids, and with this integrated general Lie algebroids. Outside diffeology, Beltita and Pelletier

[(4)] Answering a conjecture of Wockel [Woc11, Remark 7.1]

[BP26] integrated transitive Lie algebroids to their own version of "generalized smooth" groupoids. My intention is to establish the integration of transitive Lie algebroids into elastic groupoids, and thereby connect the elastic integration with Villatoro's.

Underlying Lie groupoids and Lie algebroids are singular foliations. A leaf-first approach (cf. [Miy24]) defines a singular foliation $\mathcal{F}$ as a partition of a manifold M into connected weakly-embedded submanifolds (leaves) of varying dimension, that fit together in a smooth way. Another approach (cf. [AS09], or the survey [LLR24]) defines a singular foliation as an involutive and locally finitely generated submodule $\mathcal{F}$ of compactly supported vector fields $\mathfrak{X}_c(M)$. The latter induces the former.

When $\mathcal{F}$ is a regular foliation, it integrates to the holonomy groupoid $\mathrm{Hol}(\mathcal{F}) \rightrightarrows M$. This is a Lie groupoid, and with it we may view the leaf space $M/\mathcal{F}$ as a differentiable stack. For a general singular foliation, Androulidakis and Skandalis [AS09] defined a topological groupoid $\mathrm{Hol}(\mathcal{F}) \rightrightarrows M$, which is rarely Lie. However, it carries a natural diffeology [GV22].

Question 7. In what sense does the diffeological groupoid $\mathrm{Hol}(\mathcal{F})$ differentiate to $\mathcal{F}$?

Androulidakis and Zambon gave an answer in [AZ26], but relied heavily on the existence of a functor from $\mathrm{Hol}(\mathcal{F})$ into a Lie groupoid, where the differentiation takes place. Our joint project Leonid Ryvkin and Alfonso Garmendia aims to answer this question using an intrinsic differentiation procedure. We define a class of diffeological groupoids that includes $\mathrm{Hol}(\mathcal{F})$, and that supports an appropriate Lie functor. Note that $\mathrm{Hol}(\mathcal{F})$ is not elastic in general, so we cannot use the tangent category $\mathcal{E}\mathrm{lst}$ or the tangent functor T to define this Lie functor. Instead, we directly introduce a vertical functor $\mathbb{V}\colon \mathcal{D}\mathrm{flg}^{\to} \to \mathcal{D}\mathrm{flg}$ via a left Kan extension, where $\mathcal{D}\mathrm{flg}^{\to}$ is the arrow category of $\mathcal{D}\mathrm{flg}$ (i.e., the category whose objects are smooth maps between diffeological spaces); the defining property of $\mathbb{V}$ is that, for a surjective submersion $s\colon U \to U_0$,

$$\mathbb{V}s = \mathbb{V}^s U \coloneqq \ker(Ts).$$

The primitive objects that capture our special class of groupoids, and which are compatible with $\mathbb{V}$, are *tepui*[(5)] fibrations $X \to M$, introduced in [GMR25]. The simplest of these arise as quotients of vector bundles.

Example 8. Let $V \to M$ be a vector bundle, and $D \subseteq V$ be a smooth singular subbundle.[(6)] Then $V/D \to M$ is a tepui fibration and a diffeological vector bundle, called a VB-tepui.

As a stepping-stone to integrating $\mathcal{F}$ (viewed as a submodule of $\mathfrak{X}_c(M)$), we have extended the classical Serre-Swan theorem:

Theorem 9 ([GMR25])**.** *For a fixed manifold M, the global section functor*

$$\Gamma\colon \{\textit{VB-tepui over } M\} \to \{\textit{finite-type, global, and fiber-determined } C^\infty(M)\textit{-modules}\}$$

is an equivalence of categories.

We also show that this result remains true if we add the adjective "Fréchet" to the right side, and use the functional diffeology on $\Gamma(E)$. This result assigns to $\mathcal{F}$ a canonical tepui algebroid $A_{\mathcal{F}}$. In our second forthcoming article, we integrate this tepui algebroid to $\mathrm{Hol}(\mathcal{F})$, and establish the differentiation procedure. Precisely, we show that $\mathbb{V}^{\mathbf{s}}\mathrm{Hol}(\mathcal{F}) \cong A_{\mathcal{F}}$. Our next step is to develop a theory of integration for tepui algebroids, and hence for singular foliations.

(5) Named for the distinctive South American plateau mountains.

(6) A subbundle with fibers of varying dimension

3. Riemannian diffeology

Diffeological spaces carry more than a notion of "smoothness." They also support geometric structures, such as diffeological notions of symplectic forms and Riemannian metrics. In particular, Riemannian diffeology is an active area of current research for its possible applications in infinite-dimensional settings [GW26]. Much of the framework for Riemannian diffeology was laid out in [KSS25]. The definition of a Riemannian metric on a diffeological space has two equivalent forms.

Definition 10 (Iglesias-Zemmour [Igl23])**.** A *Riemannian metric* on X is an assignment g that sends each plot $p\colon U \to X$ to a symmetric and positive covariant 2-tensor g_p on U satisfying an appropriate definiteness condition.[(7)]

Definition 11 (Kuribayashi-Sakai-Shiobara [KSS25])**.** A (weak) *Riemannian metric* on X is a smooth map $g\colon T_2X \to \mathbb{R}$ such that for each plot $p\colon U \to X$, the pullback $p^*g\colon T_2U \to \mathbb{R}$ is a symmetric and positive covariant 2-tensor. We could also impose a definiteness condition, resulting in *definite weak Riemannian metrics*.

In [Miy26a], we investigated Riemannian orbifolds. Orbifolds are naturally diffeological spaces (those locally diffeomorphic to $\mathbb{R}^n/\Gamma$, where $\Gamma \leq \mathrm{GL}(n)$ is finite), and are also naturally stacks (represented by étale proper Lie groupoids, which we assume to be effective). There is a natural quotient functor associating to a stack $\mathbb{X}$ the diffeological space $M/\mathcal{G}$, where $\mathcal{G}$ is a Lie groupoid presenting $\mathbb{X}$, whereby we can rigorously compare stacks and orbifolds. The notion of Riemannian stack was established in the papers [dF18; dF19]. We proved

Theorem 12. *The data of an effective orbifold stack $\mathbb{X}$ together with a stacky metric η is equivalent to the data of a diffeological orbifold X together with a definite weak Riemannian metric g.*

We also exhibited metrics on stacks that do not descent to metrics on the associated diffeological orbit space, illustrating that the two theories are not generally equivalent. Much work remains to advance Riemannian diffeology. Shiobara [Shi26] has established some of the fundamental theorems from Riemannian geometry (Koszul formula, Levi-Civita connection, uniqueness of geodesics) for elastic Riemannian spaces. Among my current projects is to prove a version of the Myers-Steenrod theorem, namely that for a class of elastic Riemannian spaces (X, g), such as Q-manifolds (cf. Definition 14), the isometry group is naturally a Lie group. These forthcoming results should be compared with similar ones from the theory of Riemannian stacks, such as [dd22], or from other approaches to Riemannian geometry of quotient spaces, such as [Ale+03].

4. Quasi-étale spaces and Q-manifolds

Several of the quotients above have the same property. What follows is contained in [Miy26b].

Definition 13. A smooth action of a Lie group G on a manifold M is called *lift-complete* if the only local lifts of the identity on M/G are given by the G-action. It furthermore has property (Q) if every plot $U \to M/G$ has a unique lift up to germ.

I generalize lift-completeness, and property (Q), to étale effective Lie groupoids $G \rightrightarrows M$.

Definition 14. A diffeological space is *quasi-étale* if it is isomorphic to the orbit space of some lift-complete Lie groupoid. It furthermore is a *Q-manifold* if the corresponding Lie groupoid has property (Q).

Example 15.

- With Yael Karshon [KM25], I proved that quasifolds (cf. [Pra01; IP21]) are quasi-étale. Quasifolds appear in Prato's generalization of the Delzant theorem in symplectic geometry.

[(7)] That we do not describe here.

- I proved in [Miy26b] that the leaf space of any Riemannian foliation is quasi-étale.
- With Yi Lin [LM24], we showed that most *Killing* Riemannian foliations are quasifolds. In particular, this applies to quotients G/H of Lie groups.

Lift-complete Lie groupoids and quasi-étale spaces give a setting where the higher geometric and diffeological approaches coincide.

Theorem 16. *The natural quotient functor taking a Lie groupoid to its diffeological orbit space restricts to an equivalence of: the category of étale effective lift-complete Lie groupoids, with morphisms (isomorphism classes of) surjective submersive bibundles, and the category of quasi-étale diffeological spaces, with morphisms surjective local subductions. If we impose property (Q), then we do not need to restrict the morphisms.*

This directly generalizes [KM25]. For property (Q), we have an equivalence of categories.

Question 17. Does Theorem 16 generalize to infinite-dimensional manifolds?

A positive answer would justify why the integration in Theorem 3 does not require higher structures. It would also provide a bridge to Beltita and Pelletier's work [BP25; BP24] on integrating Lie algebras and Lie algebroids, which involves infinite-dimensional variants of Q-manifolds.

References

[AB25] Aintablian, Lory and Blohmann, Christian. *Differentiable groupoid objects and their abstract Lie algebroids*. Appl. Categ. Struct. **33** (2025), no. 5. Paper 33, p. 97 (cit. on p. 3).

[Ale+03] Alekseevsky, Dmitri et al. *The Riemann geometry of orbit spaces – the metric, geodesics and integrable systems*. English. Publ. Math. Debr. **62** (2003), no. 3-4, pp. 247–276 (cit. on p. 5).

[AM85] Almeida, Rui and Molino, Pierre. *Suites d'Atiyah et feuilletages transversalement complets*. C. R. Acad. Sci. Paris Sér. I Math. **300** (1985), no. 1, pp. 13–15 (cit. on p. 3).

[AS09] Androulidakis, Iakovos and Skandalis, Georges. *The holonomy groupoid of a singular foliation*. J. Reine Angew. Math. **626** (2009), pp. 1–37 (cit. on p. 4).

[AZ26] Androulidakis, Iakovos and Zambon, Marco. *Integration of singular subalgebroids by diffeological groupoids*. English. Trans. Am. Math. Soc. **379** (2026), no. 8, pp. 5713–5763 (cit. on p. 4).

[BP24] Beltiţă, Daniel and Pelletier, Fernand. *Transitive Lie algebroids and $\boldsymbol{Q}$-manifolds*. 2024. arXiv: 2404.10607 [math.DG] (cit. on p. 6).

[BP25] Beltiţă, Daniel and Pelletier, Fernand. *Smooth Banach structure on orbit spaces and leaf spaces*. Diss. Math. **600** (2025) (cit. on p. 6).

[BP26] Beltiţă, Daniel and Pelletier, Fernand. *Transitive Lie algebroids and $\mathbf{Q}$-manifolds*. J. Geom. Phys. **228** (2026), 105919 (cit. on p. 4).

[Blo24] Blohmann, Christian. *Elastic diffeological spaces*. In: *Recent advances in diffeologies and their applications*. Vol. 794. Contemp. Math. Amer. Math. Soc., [Providence], RI, 2024, pp. 49–86 (cit. on p. 2).

[CC15] Cockett, James Robin Bernard and Cruttwell, Geoffrey S. H. *The Jacobi identity for tangent categories*. Cah. Topol. Géom. Différ. Catég. **56** (2015), no. 4, pp. 301–316 (cit. on p. 2).

[CF03] Crainic, Marius and Fernandes, Rui Loja. *Integrability of Lie brackets*. Ann. of Math. (2) **157** (2003), no. 2, pp. 575–620 (cit. on p. 3).

[dd22] del Hoyo, Matias and de Melo, Mateus. *Geodesics on Riemannian stacks*. English. Transform. Groups **27** (2022), no. 2, pp. 403–427 (cit. on p. 5).

[dF18] del Hoyo, Matias and Fernandes, Rui Loja. *Riemannian metrics on Lie groupoids*. English. J. Reine Angew. Math. **735** (2018), pp. 143–173 (cit. on p. 5).

[dF19] del Hoyo, Matias and Fernandes, Rui Loja. *Riemannian metrics on differentiable stacks*. Math. Z. **292** (2019), no. 1-2, pp. 103–132 (cit. on p. 5).

[EK64] Est, Willem T. van and Korthagen, Theodorus J. *Non-enlargible Lie algebras*. Nederl. Akad. Wet., Proc., Ser. A **67** (1964), pp. 15–31 (cit. on p. 3).

[GMR25] Garmendia, Alfonso, Miyamoto, David, and Ryvkin, Leonid. *A singular Serre-Swan theorem via tepui fibrations*. 2025. arXiv: 2510.20936 [math.DG] (cit. on p. 4).

[GV22] Garmendia, Alfonso and Villatoro, Joel. *Integration of singular foliations via paths*. English. Int. Math. Res. Not. **2022** (2022), no. 23, pp. 18401–18445 (cit. on p. 4).

[GW26] Goldammer, Nico and Welker, Kathrin. *Towards optimization techniques on diffeological spaces by generalizing Riemannian concepts*. English. Appl. Math. Optim. **93** (2026), no. 1. Id/No 5, p. 31 (cit. on p. 5).

[Igl23] Iglesias-Zemmour, Patrick. *On Riemannian metric in diffeology*. preprint. 2023 (cit. on p. 5).

[IP21] Iglesias-Zemmour, Patrick and Prato, Elisa. *Quasifolds, diffeology and noncommutative geometry*. J. Noncommut. Geom. **15** (2021), no. 2, pp. 735–759 (cit. on p. 5).

[KM25] Karshon, Yael and Miyamoto, David. *Quasifold groupoids and diffeological quasifolds*. Transform. Groups **30** (2025), no. 3, pp. 1333–1367 (cit. on pp. 5, 6).

[KM97] Kriegl, Andreas and Michor, Peter W. *The convenient setting of global analysis*. Vol. 53. Mathematical Surveys and Monographs. American Mathematical Society, Providence, RI, 1997, pp. x+618 (cit. on p. 1).

[KSS25] Kuribayashi, Katsuhiko, Sakai, Keiichi, and Shiobara, Yusuke. *Towards Riemannian diffeology*. Proceedings of the Royal Society of Edinburgh: Section A Mathematics (2025), pp. 1–30 (cit. on p. 5).

[LLR24] Laurent-Gengoux, Camille, Louis, Ruben, and Ryvkin, Leonid. *An invitation to singular foliations*. 2024. arXiv: 2407.14932 [math.DG] (cit. on p. 4).

[LM24] Lin, Yi and Miyamoto, David. *Riemannian foliations and quasifolds*. Math. Z. **308** (2024), no. 49, p. 32 (cit. on p. 6).

[Miy24] Miyamoto, David. *Singular foliations through diffeology*. In: *Recent advances in diffeologies and their applications*. Vol. 794. Contemp. Math. Amer. Math. Soc., [Providence], RI, 2024, pp. 139–160 (cit. on p. 4).

[Miy25] Miyamoto, David. *Lie algebras of quotient groups*. To appear in J. Lie Theory. 2025. arXiv: 2502.10260 [math.DG] (cit. on p. 3).

[Miy26a] Miyamoto, David. *Diffeological Riemannian orbifolds*. 2026. arXiv: 2607.08939 [math.DG] (cit. on p. 5).

[Miy26b] Miyamoto, David. *Lie groupoids determined by their orbit spaces*. J. Noncommut. Geom. (2026), 687 (cit. on pp. 5, 6).

[Pra01] Prato, Elisa. *Simple non-rational convex polytopes via symplectic geometry*. Topology **40** (2001), no. 5, pp. 961–975 (cit. on p. 5).

[Ros84] Rosický, Jiří. *Abstract tangent functors*. Diagrammes **12** (1984), JR1–JR11 (cit. on p. 1).

[Shi26] Shiobara, Yusuke. *On Riemannian Geometry in Elastic Diffeology*. 2026. arXiv: 2608.01037 [math.DG] (cit. on p. 5).

[Sou80] Souriau, Jean-Marie. *Groupes différentiels*. In: *Differential geometrical methods in mathematical physics (Proc. Conf., Aix-en-Provence/Salamanca, 1979)*. Vol. 836. Lecture Notes in Math. Springer, Berlin-New York, 1980, pp 91–128 (cit. on p. 1).

[Vil26] Villatoro, Joel. *On the integrability of Lie algebroids by diffeological spaces*. English. Indag. Math., New Ser. **37** (2026), no. 4, pp. 1192–1242 (cit. on p. 3).

[Woc11] Wockel, Christoph. *Categorified central extensions, étale Lie 2-groups and Lie's third theorem for locally exponential Lie algebras*. Adv. Math. **228** (2011), no. 4, pp. 2218–2257 (cit. on p. 3).

[WZ16] Wockel, Christoph and Zhu, Chenchang. *Integrating central extensions of Lie algebras via Lie 2-groups*. J. Eur. Math. Soc. (JEMS) **18** (2016), no. 6, pp. 1273–1320 (cit. on p. 3).